AI Enabled Smart Door for Identifying Persons with SMS Notification

An Major Project Report

Submitted to

Jawaharlal Nehru Technological University

Hyderabad

In partial fulfillment of the requirements for the

award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

By

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Approved by AICTE, New Delhi | Affiliated to JNTUH, Hyderabad | Accredited by NAAC —A‖ Grade & NBA| Hyderabad | PIN: 500068

Certificate

This is to certify that the Industry Oriented Mini Project Report on ***"AI Enabled Smart Door for Identifying Persons with SMS Notification‖*** submitted by **FAHD KHADRI** bearing HallTicketNo's **21VE1A66A0** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Artificial Intelligence & Machine Learning** from Jawaharlal Nehru Technological University, Kukatpally, Hyderabad for the academic year 2024-25 is a record of bonafide work carried out by him/her under our guidance and Supervision.

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DECLARATION

I **FAHD KHADRI** bearing Roll No's **21VE1A66A0** hereby declare that the Project titled **"*AI Enabled Smart Door for Identifying Persons with SMS Notification*‖** done by us under the guidance of **Mrs .A.SWAPNA,** which is submitted in the partial fulfillment of the requirement for the award of the B.Tech degree in **Artificial Intelligence & Machine Learning at Sreyas Institute of Engineering &** Technology for Jawaharlal Nehru Technological University, Hyderabad is our original work.

Fahd Khadri 21VE1A66A0

#Mapping of CO and PO’s of the project.

COURSE OUTCOMES OF MAJOR PROJECTS.

DOMAIN: IoT

CO1: **Identify and define** real-world problems that can be effectively addressed using IoT-based solutions, considering sensor integration and connectivity requirements.

CO2: **Design** an IoT architecture incorporating appropriate hardware components (sensors, microcontrollers, and actuators) and communication protocols (e.g., MQTT, HTTP, Zig Bee).

CO3: **Develop and implement** functional IoT prototypes using platforms such as Arduino, Raspberry Pi, or ESP32, and interface them with cloud services or mobile/web applications.

CO4: **Analyse and evaluate** the performance of the IoT system in terms of data accuracy, latency, power consumption, and scalability.

CO5: **Demonstrate** effective time and resource management, and awareness of security, privacy, and ethical implications in IoT deployments.

CO6: **Collaborate** to integrate individual contributions into a team effort and complete the design.

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	✓	✓				✓	✓	✓					✓	✓
CO2	✓	✓	✓	✓	✓		✓				✓		✓	✓
CO3	✓		✓	✓	✓				✓	✓			✓	✓
CO4	✓		✓	✓	✓							✓	✓	✓
CO5		✓				✓	✓	✓			✓	✓		
CO6			✓						✓	✓	✓			

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Fahd Khadri 21VE1A66A0

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ABSTRACT

This paper presents an advanced smart door security system which uses artificial intelligence to identify the person accurately and send immediate SMS-based notifications. Using an ESP32-CAM module along with a specially tuned convolutional neural network (CNN) model for building a vision-based recognition mechanism, the proposed architecture overcomes some of the primary limitations of conventional sensor-based approaches. The architecture system incorporates a SIM800C GSM module for robust wireless connectivity along with an Arduino Uno microcontroller for real-time processing. The technology scans immediately the facial characteristics of an individual, verifies his/her identity against pre-stored patterns, and utilizes the GSM network to notify through SMS the pre-registered phone numbers as it detects them drawing near to the door. A buzzer provides immediate on-site notification, and a special 16x2 LCD displays local status messages. A constant 12V DC power supply provides energy to the hardware which ensures continuous operation. unlike traditional PIR sensor systems, which can tend to generate false alarms, this method demonstrates good accuracy over a variety of viewing in different angles and lighting conditions. The effectiveness of the system in real-world scenarios is validated by experimenting it in all the real world conditions , which indicates virtually instant SMS delivery and over 90% successful recognition rates. By providing a low-cost, microcontroller-based system that integrates computer vision and wireless communication, this work promotes forward intelligent security technology. The system has valuable applications in surveillance of restricted areas, management of office access, and home security. Future enhancements could consist of multi-factor authentication for more security and cloud connectivity for monitoring remotely. The research demonstrates the way in which state-of-the-art AI technology can be employed effectively to smart security solutions using affordably priced hardware components.

Keywords: Artificial intelligence, face recognition, smart security system, embedded systems, IoT, wireless notification, access control, microcontroller.

CHAPTER 1

INTRODUCTION

Traditional door security systems which involves PIR sensors, mechanical locks, and keypads are also had many disadvantages, including chances to unauthorized entry, high false alarm rates, and faulty provisions for real-time monitoring. There is an immediate need for more advanced solutions that can provide advanced safety without compromising user convenience in view of these needs. The integration of sophisticated but cost-effective security systems with automatic identification and real-time notification is possible today due to recent progress in artificial
Intelligence and embedded systems.

This work introduces a new smart door system that combines real-time SMS alerting and facial recognition using computer vision Basically, the system employs an ESP32-CAM module with a CNN model, an Arduino Uno microcontroller to process images, and a SIM800C GSM module for communication. This setup is preferable compared to traditional security measures because it offers some distinguishing advantages such as enhanced detection of humans, fewer false alarms, and real-time alerting possibilities. Vision-based recognition is better than ordinary issues with the conventional sensor-based approaches as it functions under fluctuating lighting and observation angles. The in-field demonstration demonstrates how this hybrid solution can be utilized in an array of environments, from high-security facilities to residential homes.

Standard door security systems with PIR sensors, mechanical locks, and keypads are also riddled with many disadvantages such as vulnerability to illegal access, excessive false alarms, and lack of features for real-time monitoring. More innovative solutions which can deliver reliable safety solutions without compromising user convenience are in high demand in due to these limitations.

Performance testing verifies the system’s ability in actual real world environments, highlighting the system's particular strengths for access control applications. We have designed a solution that is both technologically advanced and cost-effective to implement by integrating wireless notification and facial recognition technologies into one embedded platform. Future enhancements and customization according to specific security needs are also feasible due to the modular design with improved performance and a cost-effective alternative to expensive commercial solutions, this work represents a significant advance in smart security systems. The frictionless integration of productive IoT applications and identification via AI opens up intelligent deployment opportunities. In addition to the immediate applications, the system design builds a framework on which access control technology will further develop and in more complex ways subsequently with potentially additional biometric authentications and more advanced threat modelling in future releases.

1.1 Problem Statement

The growing number of thefts, unauthorized access, and breaches in residential areas calls for enhanced home security systems that offer real-time, intelligent surveillance. Traditional door locking mechanisms such as manual locks and keys, RFID access cards, and numeric keypads have become outdated due to their limited functionalities and vulnerability to security compromises. RFID systems, for example, depend on physical tokens that can be lost, stolen, or duplicated. Similarly, keypad entry systems are prone to being hacked or guessed. These methods are reactive rather than proactive and do not provide homeowners with real-time alerts or situational awareness.

CCTV systems, although widely used, have their limitations. They offer passive surveillance, capturing footage without active monitoring or instant alerts. A homeowner can only review footage after an incident has occurred. The delay in response time can lead to significant losses or threats to safety. In the absence of automation and remote notification, traditional systems do little to deter intruders or inform the homeowners of any breach.

Additionally, these systems lack adaptability. They cannot learn from patterns or adjust their behavior according to different situations. For example, recognizing a frequent visitor versus an unknown individual requires manual observation or intervention. This gap necessitates a smart solution that integrates AI for intelligent recognition, IoT for connectivity, and communication tools like GSM for real-time alerts.

Therefore, there is an urgent requirement for a system that not only secures entry points but also identifies, monitors, and alerts homeowners in real-time using advanced AI algorithms and communication protocols. The system should be able to detect, recognize, and respond autonomously to ensure the safety and security of residents.

1.1.1 Objectives:

The primary goal of this project is to design and implement a smart, AI-driven door security system that utilizes facial recognition to identify individuals and sends real-time SMS notifications to homeowners upon detection. The proposed system is aimed at achieving the following objectives:

This project aims to revolutionize residential and small-scale commercial security by integrating advanced technologies into a unified, intelligent system. At its core, it employs facial recognition powered by Convolutional Neural Networks (CNNs), allowing the system to accurately identify and authenticate individuals based on facial features. By using CNNs, the system benefits from high precision in recognizing faces under varying lighting conditions and angles, thus improving reliability and minimizing the chances of unauthorized access.

To enable real-time video capture and processing, the ESP32-CAM module is utilized. This low-cost, Wi-Fi-enabled microcontroller offers onboard camera support, allowing the system to capture live video streams and analyze them on the edge without depending on external servers or cloud processing. This not only ensures faster response times but also enhances data privacy, as sensitive visual information is processed locally. The ESP32-CAM is seamlessly interfaced with the Arduino Uno, which handles peripheral I/O operations such as controlling locks, triggering alarms, and coordinating data between different modules, ensuring harmonious system operation.

In addition to facial recognition and video processing, the system incorporates the SIM800C GSM module to maintain robust communication capabilities. When a face is detected, especially if it is unrecognized or unauthorized, the module sends instant SMS alerts to pre-registered mobile numbers, notifying homeowners or security personnel of potential intrusions. This real-time alert mechanism enables prompt responses, even when the user is far from home.

The project eliminates the need for conventional access control mechanisms such as keys, PIN codes, or RFID cards, which can be lost, stolen, or duplicated. Instead, it employs AI-driven decision-making that autonomously identifies individuals and determines access rights. This automation not only enhances convenience but also reduces human error and dependency on physical credentials.

To address common issues like false alarms caused by lighting changes, pets, or unfamiliar objects, the system is fine-tuned with advanced face verification techniques and adaptive thresholding. Only confirmed matches are allowed to trigger access, thereby maintaining both high security and user satisfaction. The system's compact and modular design makes it ideal for a variety of environments—from private residences to office entrances and restricted areas—while remaining cost-effective and easy to deploy.

Furthermore, by offering continuous surveillance and the ability to respond in real time, the system significantly enhances situational awareness. Homeowners can rest assured that their property is under 24/7 intelligent monitoring, providing peace of mind and a proactive layer of defense against intrusions. In future iterations, the project can be extended to include cloud-based dashboards, mobile apps for real-time video feeds, and voice control, making it a fully integrated smart security solution.

CHAPTER 2

LITERATURE SURVEY

In the past few years, there has been a significant need for secure and smart home automation systems power by urbanization, security needs, and the availability of low-cost embedded computing platforms. Door security systems becoming essential because they represent the first point of defence against robbers. The integration of embedded hardware, artificial intelligence (AI), and wireless communication modules has enabled smarter, faster, and more efficient monitoring and control mechanisms for both residential and commercial setups.

However, a thorough examination of existing door security systems reveals that most implementations still rely on traditional control mechanisms such as RFID, keypad entry, motion sensors, or simple face recognition algorithms with limited AI capabilities. Further, some of these models do not incorporate real-time alert features or remote access features, which are essential to achieving instant responses in case of security breaches. This section criticizes some of the notable contributions in the area, their respective methodologies, and the current limitations, providing the basis for the proposed improvement—a GSM-based door security system powered by AI.

Haque et al. [1] proposed a face recognition-based door lock system using OpenCV and Arduino Uno. While this system provided a basic level of biometric verification, it lacked integration with GSM modules for real-time alerting. The system functioned effectively at the local level but did not support remote communication or advanced AI-based decision- making.

Kassim et al. [2] developed an RFID-based access control system using Java. The solution provided a fast and contactless authentication process; however, it was vulnerable to RFID tag cloning and did not support biometric identification. Its reliance on physical tokens made it susceptible to unauthorized access through lost or replicated cards.

Bhat et al. [3] combined RFID and GSM technologies using NodeMCU for attendance automation and room security. Despite incorporating wireless communication for status updates, the system lacked facial or biometric authentication, depending entirely on card-based validation. This posed risks related to card duplication or misuse.

Hadi et al. [4] introduced a smart home security framework integrating IoT modules with a Network Intrusion Detection System (NIDS). Although it offered SMS-based alerts for security events, it lacked facial recognition or biometric capabilities, and alert reliability was impacted by mobile network delays.

Frimpong [5] used the Face++ cloud API to implement a smart burglar alarm system via Arduino. The cloud-based facial recognition improved accuracy but introduced latency and dependence on third-party servers. This compromised performance in low-connectivity environments and raised concerns regarding data security.

Chandra [6] implemented a facial recognition system using ESP32-CAM and a relay module for door locking. While it successfully operated on local networks with embedded AI capabilities, the absence of GSM alerts limited its effectiveness in notifying homeowners during unauthorized access when they are off-site.

Golder [7] designed a GSM-based home security system employing infrared (IR) sensors for intrusion detection. Although the system could send alerts via GSM, it did not differentiate between users, lacking facial or biometric identification features. It was therefore unsuitable for authentication-critical scenarios.

Ouni [8] proposed an AIoT system integrating Maixduino microcontrollers with GSM and edge AI inference. Despite demonstrating potential in edge-based recognition, the system struggled with processing power limitations and unstable network performance during field deployments.

Abdulsalam [9] developed a GSM-based door lock using Arduino and relay modules. This solution enabled remote access via SMS but offered no facial or biometric recognition. The system operated blindly to the identity of the person requesting access.

Debnath [10] introduced a Wi-Fi-enabled smart door lock using the ESP8266 microcontroller and the Blynk app. The system allowed remote operation but lacked biometric authentication or GSM support, making it less robust for remote or offline conditions.

Hasan et al. [11] demonstrated facial recognition using OpenCV and a laptop webcam. While the model provided accurate recognition, it was not embedded and lacked

GSM or alerting capabilities. It was primarily a prototype rather than a deployable security solution.

Norarzemi et al. [12] presented an IoT-based door lock system using NodeMCU and the Blynk platform. Although easy to implement, it lacked intelligent decision-making or facial recognition, relying on simple command-response actions.

Amesimenu et al. [13] built a Bluetooth-controlled door lock system using the HC-05 module and Arduino. Limited by range and connectivity, this system did not incorporate any Intelligent user recognition, restricting its application to very localized areas.

Ihsan et al. [14] introduced a PIR and GSM-based system for motion detection. It could send alerts upon detecting movement but could not identify or differentiate between individuals, often resulting in false positives.

Orji et al. [15] implemented an RFID-based access system using Arduino. Although functional, it lacked security due to the susceptibility of RFID cards to spoofing and did not offer biometric or smart features.

Santhosh et al. [16] created a voice-controlled locking system. The solution failed in noisy environments and lacked fallback authentication methods, visual recognition, or alerting mechanisms.

Surantha et al. [17] built a motion-triggered alarm system using PIR sensors and buzzers. While effective in detecting presence, the system was highly sensitive, generating frequent false alarms and offering no recognition features.

CLAUDINE et al. [18] introduced a keypad-based door lock using Arduino UNO. It provided basic password protection but was vulnerable to brute-force attempts and lacked notification or biometric features.

Kader et al. [19] implemented a biometric fingerprint-based lock using Arduino. Though it improved security via fingerprint verification, it lacked GSM alerting and remote accessibility, making it less adaptable to unattended conditions.

2.1 Existing System

Existing door security systems generally fall into several categories: RFID-based, PIN/password-protected, facial recognition-based systems without AI, and systems that use GSM modules for SMS alerts. Each system has strengths but also clear weaknesses. RFID-based systems, while widely used, are vulnerable to cloning and unauthorized access if cards are lost or duplicated. Password-based locks can be guessed or shared and offer no biometric or contextual user authentication. Basic facial recognition systems often rely on cloud APIs, introducing latency and data privacy concerns.

For instance, Arduino-based systems are widely deployed due to their ease of use and support for sensors. However, most of these implementations perform rudimentary checks and lack advanced biometric authentication, such as real-time facial recognition powered by machine learning models. Many systems use PIR sensors or motion detection for presence sensing, which often produce false positives or lack identity verification. GSM-enabled systems can send alerts but lack the intelligence to determine who is at the door.

A few recent works have attempted to use ESP32-CAM for facial recognition in embedded environments, but these models operate solely on local networks and cannot provide remote alerting. Also, most implementations do not integrate deep learning or AI models like CNNs for image classification, nor do they use pre-trained models for efficient identification. These systems work well for basic identification but fail in terms of security, automation, and real-time communication.

Despite several advancements in sensors and embedded computing, many door automation systems today lack a unified, intelligent mechanism that includes both recognition and notification capabilities. Current systems are either standalone facial recognition or GSM-only SMS notification systems but not a combined architecture. Additionally, many fail to adapt to variable lighting conditions or environmental changes, limiting their reliability.

2.1.1 Drawbacks of Existing Systems

i. **Lack of Real-Time AI Recognition:** Many systems either rely on cloud-based APIs or simple image comparison techniques, which do not scale well for dynamic lighting, camera angles, or facial expressions. These limitations hinder their usability in real-world scenarios.

ii. **No Unified System:** Most solutions either implement GSM for notification or facial recognition, not both. Systems with only one feature lack comprehensive security. Users cannot both verify and respond remotely.

iii. **Poor Internet Dependency:** Cloud-based systems need stable internet connectivity. In areas with poor internet service, these systems become ineffective or unusable, defeating the purpose of remote alerts.

iv. **Vulnerability to Physical Attacks:** RFID and keypad-based models are prone to spoofing or brute-force attacks. They offer no layered or multi-modal security such as combining face, time of access, and alerts.

v. **Inability to Learn:** Most systems lack adaptive intelligence. They cannot improve with usage or recognize familiar visitors more efficiently over time. Without machine learning or training feedback, systems remain static and less secure.

vi. **No Scalability:** Adding more features, like multi-user access or remote locking, is difficult in basic Arduino-based models, which lack memory and processing power. Legacy systems cannot be upgraded with AI features without major overhauls.

2.2 Proposed System

The proposed AI-powered smart door system offers a next-generation approach to enhancing residential security by merging real-time computer vision, embedded intelligence, and rule-driven automation. This innovative system eliminates the dependency on traditional keys, RFID cards, and manually operated locks by integrating advanced artificial intelligence (AI) and machine learning (ML) techniques. The door is capable of independently recognizing individuals, making decisions based on facial authentication, and sending instant alerts to homeowners through SMS communication. The seamless interplay of detection, recognition, and decision logic forms a robust security framework suited for modern smart homes.

Unlike conventional systems that depend on cloud processing or external verification, our solution processes data locally using edge computing. This reduces latency, enhances responsiveness, and eliminates the risk of internet downtime. It ensures privacy, as sensitive facial data does not need to be uploaded to third-party servers. The system intelligently distinguishes authorized individuals from unknown faces, initiates appropriate access control actions, and notifies the user in real-time. This is achieved through the coordinated use of three primary technical strategies: face detection using the Haar Cascade Algorithm, facial recognition using the Local Binary Pattern Histogram (LBPH) method, and decision-making via Rule-Based AI Logic.

2.2.1 Face Detection using Haar Cascade Algorithm

The first layer of intelligence in the smart door system is responsible for detecting human faces from live video input. The Haar Cascade Classifier is employed here, a supervised learning algorithm widely recognized for its speed and accuracy in object detection. It operates by scanning the image at multiple scales and positions to identify facial features based on predefined patterns trained from datasets of positive (face) and negative (non-face) images.

The camera continuously captures frames which are converted into grayscale to simplify computation. The algorithm performs a sliding window search over these frames, extracting Haar-like features at each step. These features represent contrasts in adjacent rectangular regions (such as eyes and nose) which are characteristic of human faces. These are evaluated using a cascade of increasingly complex classifiers. If the image passes through all stages of the classifier, a face is said to be detected.

This detection mechanism ensures the system activates only when a human face is identified in the camera feed, conserving energy and computation. Its effectiveness in varied lighting and environmental conditions makes it ideal for real-time, embedded door systems.

2.2.2 Face Recognition using Local Binary Pattern Histogram (LBPH)

After detecting the presence of a human face, the next step is to recognize whether the individual is authorized. For this, the system uses the LBPH algorithm, which is computationally efficient and robust in handling diverse lighting conditions. LBPH analyzes local texture patterns of a face by comparing each pixel with its surrounding neighbors. This comparison yields a binary pattern, which is then converted into a decimal value, forming a histogram representing specific facial characteristics.

The entire facial image is divided into a grid of small regions, and histograms are calculated for each. These local histograms are concatenated to form a global representation of the face. During training, these histograms are stored in a database against known identities. During real-time execution, the incoming face's histogram is compared to those in the database using Euclidean or chi-square distance metrics. The closest match confirms identity, while no match triggers an unrecognized person alert.

LBPH stands out for its high accuracy, simplicity, and resistance to variation in light or orientation, making it the ideal choice for embedded smart door systems with limited processing power.

2.2.3 Decision Control using Rule-Based AI Logic

Once a face is recognized or flagged as unknown, the system must determine what action to take. This is governed by a rule-based artificial intelligence logic engine. Unlike black-box models, rule-based AI is transparent, allowing users to define and modify access behavior based on specific conditions.

The control logic evaluates multiple inputs—such as the result of face recognition, time of access, or even previous access logs—to decide the next action. If a person is authenticated, the system sends an SMS confirmation and activates the door-unlocking mechanism. If the face is unknown or suspicious, it immediately triggers a buzzer and sends an alert SMS to the homeowner. Furthermore, if the camera detects movement without identifying a face, it may also trigger a motion-based alert, providing proactive surveillance.

The flexibility of this logic layer allows for easy scaling. For example, users can set time-based access rules, disable entry outside working hours, or integrate additional sensors.

The deterministic nature of rule-based logic ensures predictability and accountability—crucial for any security-sensitive application.

By integrating HaarCascade detection, LBPH recognition, and rule-driven control, the proposed smart door system delivers an efficient, secure, and highly responsive experience. The local processing capability ensures speed and privacy, while the modular design supports future enhancements, such as voice control or multi-factor authentication. Overall, this intelligent security system addresses the evolving needs of modern homeowners seeking automation without compromising safety.

CHAPTER 3
SYSTEM DESIGN

3.1 Importance of the Design

The design of the AI-enabled smart door system plays a pivotal role in delivering a secure, intelligent, and real-time access control mechanism that is adaptable to dynamic environments. The architecture is thoughtfully structured to integrate vision-based detection, AI-driven recognition, and autonomous decision-making, ensuring seamless user interaction and optimal system functionality. The design emphasizes local edge processing for real-time responsiveness, privacy, and independence from cloud-based dependencies.

The system initiates with the continuous capture of image data through the ESP32-CAM module, which operates as the primary sensory unit. The facial detection is efficiently handled using the Haar Cascade classifier algorithm, selected for its ability to rapidly detect human faces with minimal computational overhead. Once a face is detected, the image undergoes preprocessing, including grayscale conversion and normalization, to optimize it for the subsequent recognition stage.

Facial recognition is accomplished using the Local Binary Pattern Histogram (LBPH) algorithm, which is both lightweight and robust in performance. It processes texture patterns from the detected face and compares them with stored profiles in the database to authenticate individuals. The decision engine, powered by rule-based AI logic, interprets the recognition outcome to determine the appropriate action—unlocking the door, triggering a buzzer, or sending an SMS alert via the SIM800C GSM module. These core components are integrated into a unified system controlled by the Arduino Uno, which coordinates sensor inputs and output responses.

The importance of the design lies in its ability to function autonomously. Unlike cloud-based systems that depend on internet availability, this design ensures that facial detection and recognition are performed locally, enabling uninterrupted performance even in offline environments. This local processing reduces latency significantly, making the system highly suitable for real-time applications. Furthermore, it eliminates potential privacy issues by keeping sensitive facial data within the device.

The user interface, though minimal in this system, is designed to provide necessary real-time feedback through an LCD display and buzzer alerts. These elements enhance usability by keeping users informed of system actions such as access granted, access denied, or alert triggered. Additionally, the system is designed for future scalability. New facial data can be added to the training dataset, allowing the model to evolve as more individuals are registered.

The design also incorporates modularity, which enables each component—face detection, recognition, decision logic, and communication—to be independently upgraded or replaced without affecting the entire system. This modular approach enhances maintainability and allows the integration of additional security layers like fingerprint scanners or two-factor authentication if required in the future.

Security is further reinforced by integrating logic that logs unsuccessful access attempts and unauthorized detection events. These logs can be stored locally and retrieved when needed for auditing. In deployment scenarios such as gated communities, schools, or office premises, the design allows for centralized alert management where multiple doors can be monitored from a single dashboard.

Overall, the system design underscores not only technical sophistication but also practical deployment feasibility. By balancing performance, cost-efficiency, and scalability, the smart door system meets the core objectives of modern intelligent access control solutions. The emphasis on local processing, real-time decision-making, and automation ensures the system is both responsive and secure, while its adaptable architecture makes it a strong foundation for future advancements in smart home security.

3.2 UML Diagrams

3.2.1 Use Case Diagram

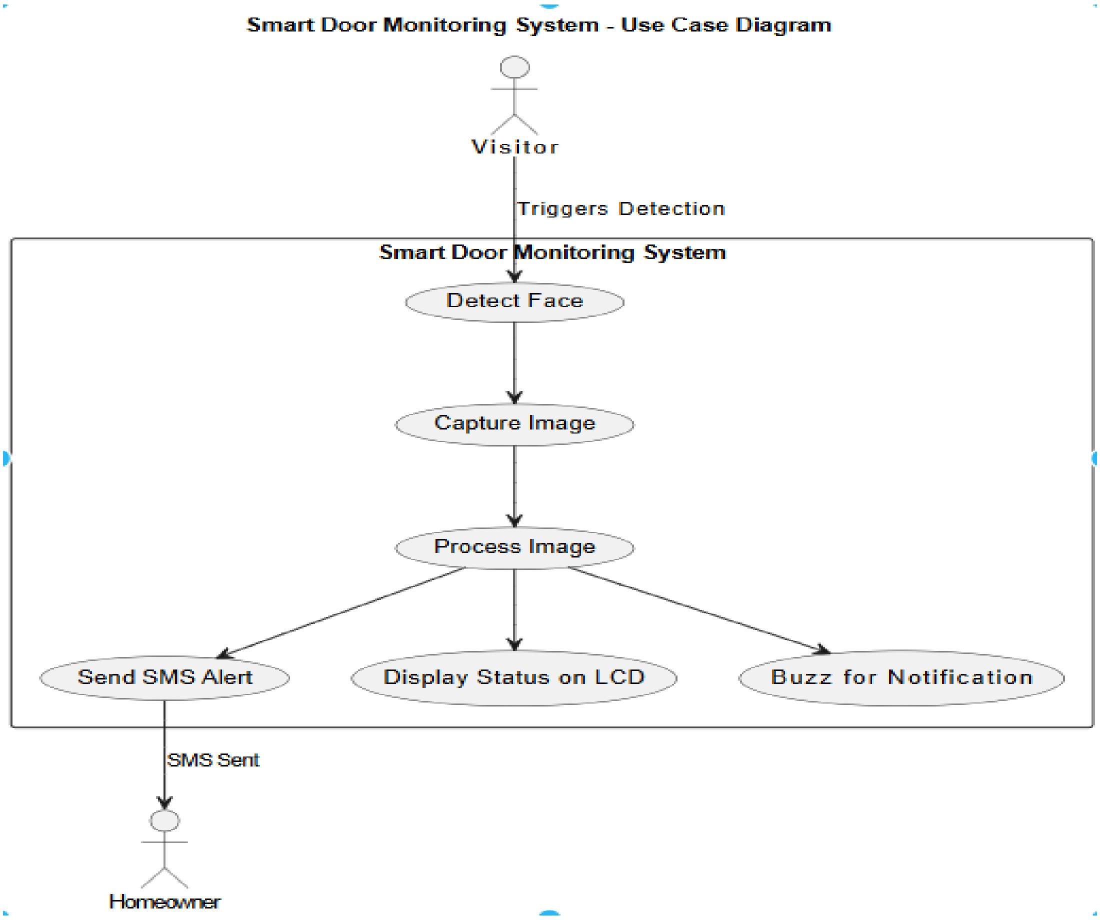

Fig 3.2.1: Use Case Diagram for AI-enabled smart door monitoring system sarcasm detection and sentiment analysis

The use case diagram for the AI-enabled smart door monitoring system provides a visual overview of how various entities interact with the system to fulfill specific functional requirements. At the center of this interaction is the visitor, who triggers the entire process by approaching the smart door. This action sets off a sequence of use cases within the system that collectively enable secure and intelligent door monitoring and notification.

Upon the visitor's presence, the system initiates face detection, which is responsible for locating facial features in the captured video feed. Once a face is detected, the next step is to capture an image of the visitor. This captured image is then processed internally by the system to identify the individual. The internal processing includes facial recognition and classification, where the system compares the detected face against stored profiles using pre-trained algorithms such as LBPH.

Depending on the recognition outcome, the system performs different actions. If the person is recognized as an authorized individual, the system may unlock the door and display a status message on the LCD. If the person is unrecognized, the system will buzz to alert those inside the premises and simultaneously trigger the SIM800C GSM module to send an SMS notification to the homeowner.

This use case diagram not only captures the flow of actions triggered by a visitor but also highlights how the system autonomously handles recognition, decision-making, and notification. The homeowner, as a secondary actor, is informed through SMS and is not required to be physically present for verification or access control. This facilitates remote monitoring and enhances home security.

The modular structure of the diagram reflects the scalable nature of the system. Additional functionalities such as data logging, facial database updates, or integration with smart home ecosystems can be added as new use cases. The clear division of responsibilities between actors and the system enables intuitive understanding of operational logic, making the diagram a critical component in the design and documentation of this project.

3.2.2 Sequence Diagram

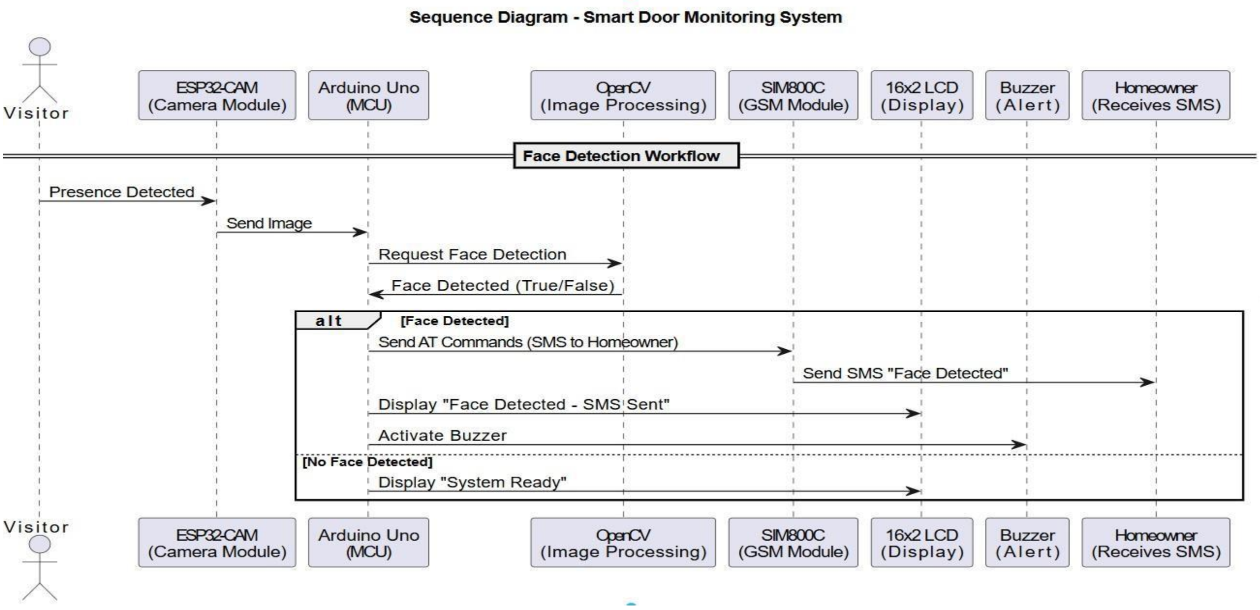

Fig 3.2.2: Sequence Diagram for AI-enabled smart door monitoring system sarcasm detection and sentiment analysis

The sequence diagram for the Smart Door Monitoring System visually represents the dynamic workflow of an AI-enabled door security system designed to identify individuals using face recognition and to notify homeowners via SMS. According to the project documentation, this system primarily aims to enhance home security using a vision-based approach, as opposed to traditional methods like RFID cards or motion sensors, which lack accuracy and contextual analysis.

The process begins when the ESP32-CAM camera module detects the presence of a visitor at the doorstep. Once a human presence is identified, the camera captures an image and transmits it to the Arduino Uno microcontroller. The Arduino then forwards a request to the OpenCV-based image processing unit, which utilizes Convolutional Neural Networks (CNNs) to perform face recognition. These CNN models are trained on diverse datasets to accurately detect and identify human faces under various lighting conditions and angles.

If a face is detected in the image, the Arduino proceeds to send AT commands to the SIM800C GSM module, prompting it to dispatch an SMS alert to the registered homeowner's mobile number. At the same time, the LCD display is updated with a message indicating that a face has been detected and that an SMS notification has been sent. Additionally, a buzzer is activated to audibly signal that someone is at the door. This immediate multi-sensory feedback ensures that any authorized or unauthorized access attempt is noticed in real time.

However, if no face is detected in the captured image, the system takes no further action besides updating the LCD display to show a "System Ready" status. This prevents unnecessary alerts or disturbances due to false detections caused by objects, animals, or shadows.

Overall, the system demonstrates an intelligent interaction between hardware components like ESP32-CAM, Arduino Uno, GSM module, and software modules leveraging OpenCV for AI-based face recognition. It efficiently integrates presence detection, image capture, AI processing, conditional SMS notification, and user interface alerts, all tailored to provide a robust and responsive home security solution.

3.2.3 Activity Diagram

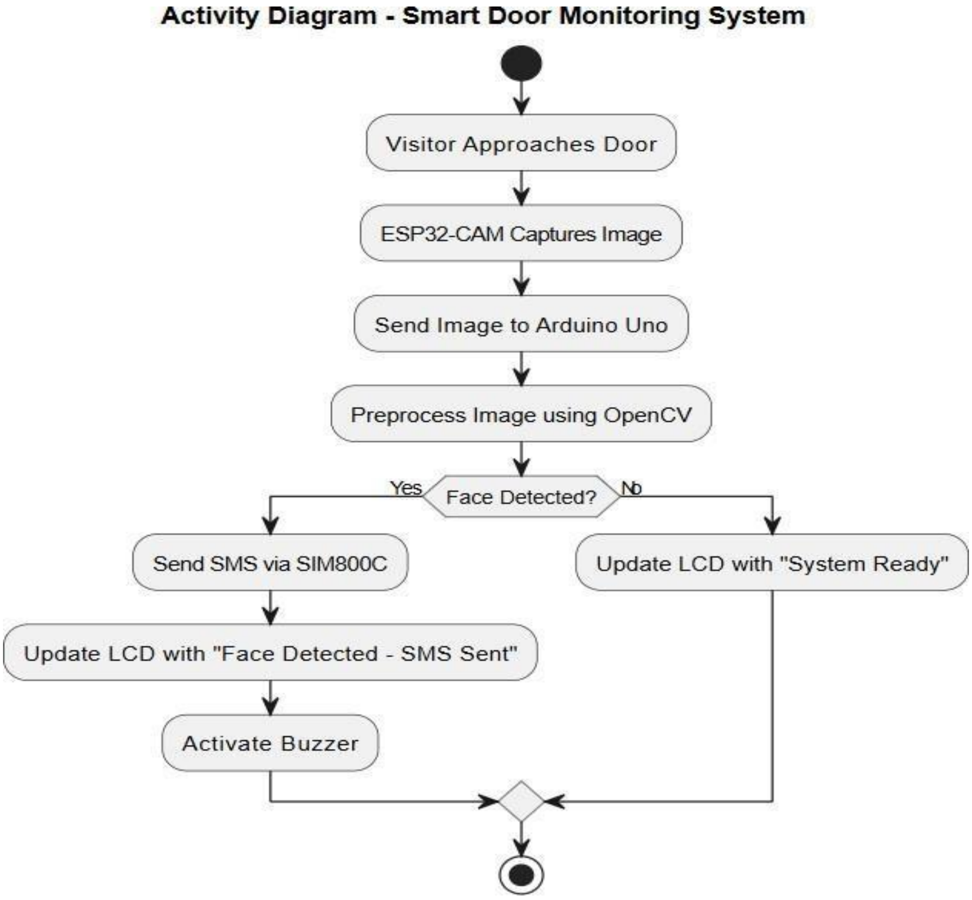

Fig 3.2.3: Activity Diagram for AI-enabled smart door monitoring system sarcasm detection and sentiment analysis

The activity diagram for the AI-enabled Smart Door Monitoring System presents a step-by-step visual representation of how the system behaves when a visitor approaches the door. It highlights the logical flow of activities involved in detecting human presence, performing facial recognition, and subsequently alerting the homeowner, encapsulating the system's behavior over time. As a behavioral UML diagram, it emphasizes the sequence of operations and decision-making pathways that the system undertakes, making it essential for understanding and modeling dynamic aspects of the door monitoring workflow.

The process initiates when a visitor comes near the smart door. This triggers the ESP32-CAM module to capture an image of the visitor. The image is then transmitted to the Arduino Uno microcontroller, which plays a central role in managing data flow and interactions between different components. After receiving the image, the Arduino directs the image to the OpenCV framework for preprocessing and analysis. Here, OpenCV, in conjunction with a pre-trained Convolutional Neural Network (CNN), is responsible for detecting whether a human face is present in the image.

The core decision point in the workflow is the face detection check. If a face is successfully detected, the system proceeds to send an SMS notification to the registered homeowner via the SIM800C GSM module. This action is immediately followed by an update on the LCD screen, which displays the message "Face Detected - SMS Sent" to provide a visual confirmation. To further alert the household of a visitor's presence, the buzzer is activated, offering an audio cue that someone is at the door.

On the other hand, if no face is detected during the image processing stage, the system follows an alternative path. Instead of triggering the GSM module or buzzer, it updates the LCD with the message "System Ready," indicating that the system is operational but no actionable visitor interaction has occurred. Both of these branches eventually converge at the end node, completing one full cycle of the monitoring process.

Overall, this activity diagram reflects a structured and automated approach to smart home security. It shows how various components—camera, microcontroller, image processing, and communication modules—work in a synchronized manner. The diagram also clearly visualizes conditional paths, emphasizing the decision-making logic that determines whether an alert is sent or the system returns to standby. It effectively conveys the dynamic behavior of the smart door system, offering a high-level yet precise overview suitable for both technical understanding and system design validation.

3.2.4 System Architecture

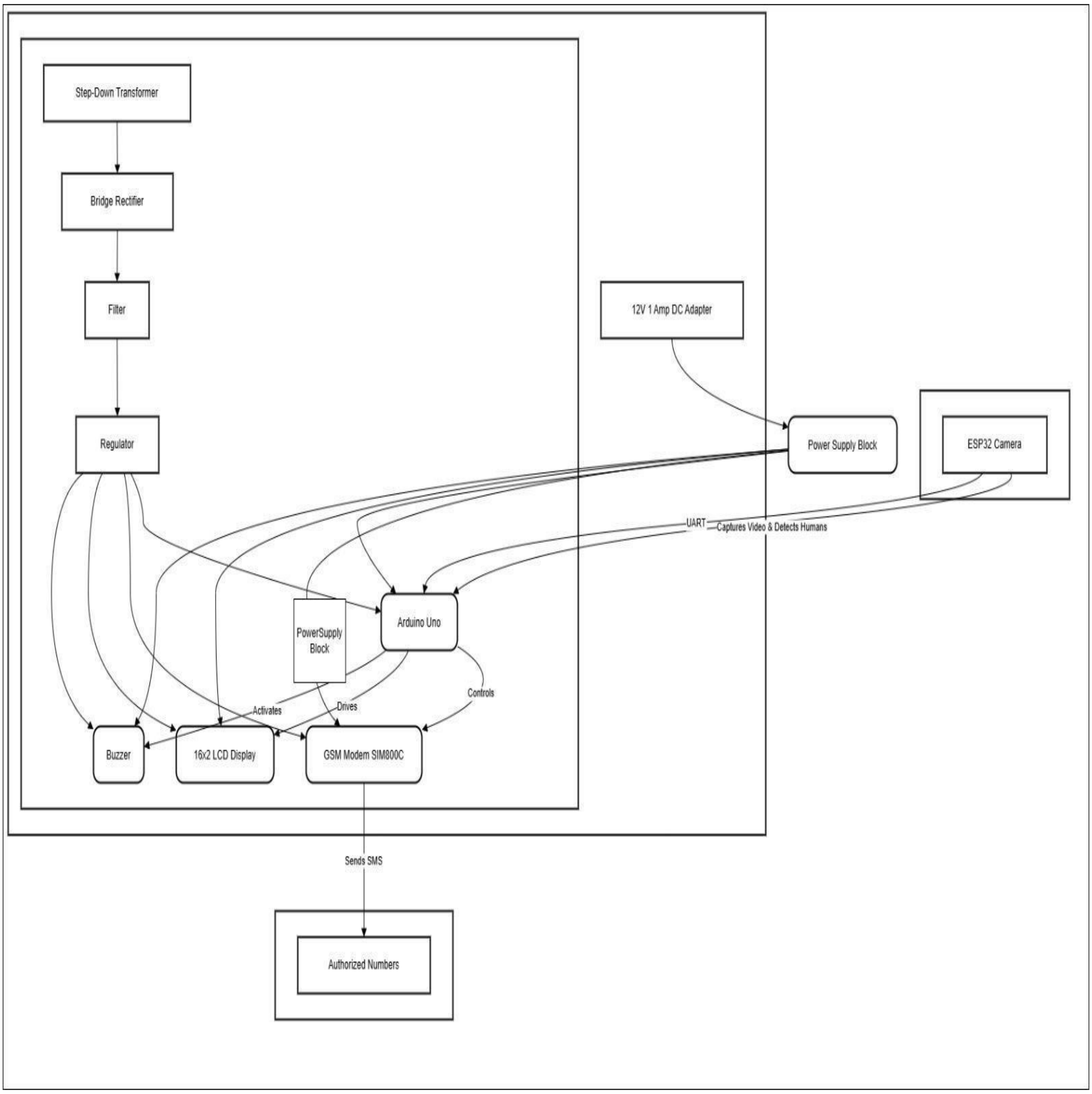

Fig 3.2.4: System Architecture for AI-enabled smart door monitoring system sarcasm detection and sentiment analysis

The block diagram of the AI-enabled Smart Door Monitoring System provides a comprehensive overview of the hardware architecture and how each component is functionally interconnected to perform intelligent surveillance and alert operations. At the heart of the system lies the Arduino Uno microcontroller, which serves as the central control unit. It interfaces with multiple peripheral devices including the ESP32 camera, GSM modem

(SIM800C), 16x2 LCD display, and a buzzer, orchestrating the system's logical and physical responses.

The power management starts from a step-down transformer, which reduces the incoming AC voltage to a manageable level. This is followed by a bridge rectifier that converts AC to DC, a filter to smoothen the DC output, and a voltage regulator that ensures a stable voltage supply to all components. A parallel power pathway is shown leading to a 12V 1 Amp DC adapter, feeding into a power supply block that powers the ESP32-CAM, allowing it to function independently yet remain in communication with the Arduino Uno via UART. The ESP32-CAM is responsible for capturing video streams and detecting human presence, initiating the entire monitoring process.

When a person is detected at the door, the ESP32-CAM sends a signal to the Arduino Uno, which processes the input and determines whether to trigger the alert mechanism. The GSM modem (SIM800C), driven by the Arduino, sends SMS notifications to a preconfigured list of authorized numbers. Simultaneously, the LCD display is updated with real-time system status messages, and the buzzer is activated to provide an audible alert. Each of these devices draws power from the regulated supply line, and their operations are finely coordinated through control signals from the Arduino.

This hardware block diagram illustrates a robust and scalable design for smart security. It balances autonomy with control, allowing real-time detection, recognition, and communication. With clear demarcation of power flows and control lines, the system demonstrates how hardware integration supports intelligent, AI-powered surveillance at the edge, enabling homeowners to receive prompt alerts and monitor their front door with enhanced reliability and automation.

3.3 Functional Requirements

3.3.1 Software Requirements:

- **Development Environment & Tools:**
  The system requires Windows 10 or higher (or Ubuntu 20.04+), and development can be carried out in Jupyter Notebook, VS Code, or PyCharm using Python 3.8 or above. Arduino IDE is used for programming microcontrollers.

- **Frameworks & Libraries:**
  Core technologies include TensorFlow or PyTorch for implementing AI models (e.g., face recognition), OpenCV for image processing, and dlib for facial detection. Supporting libraries such as Transformers, NumPy, Pandas, Matplotlib, and Seaborn are also required, along with pip or conda for dependency management.
- **Functionality & Integration Constraints:**
  The software must support multitask learning for real-time face detection and classification, integrate pre-trained CNN/BERT models, and ensure efficient handling of large datasets and low-latency communication with hardware components like GSM and LCD modules.

3.3.2 Hardware Requirements:

- **Core Components & Processing Unit:**
  The system is built around an Arduino Uno or Mega microcontroller, interfacing with an ESP32-CAM for real-time video capture. For enhanced capabilities, a Raspberry Pi may be used for local AI inference and processing.
- **Peripheral Devices:**
  Includes a SIM800C or SIM900A GSM module for SMS alerts, a 16x2 LCD with I2C for displaying system status, and a 5V piezo buzzer for audible notifications. A PIR sensor can be optionally added for initial motion detection.
- **Power & Connectivity:**
  A 12V 1A DC adapter supplies power, regulated by components like the LM7805. Communication between modules is managed via UART, with Wi-Fi support through the ESP32-CAM for optional remote access.
- **Storage & Performance Needs:**
  A minimum of 8 GB RAM (16 GB recommended), Intel i5 or AMD Ryzen 5 processor, and at least 250 GB of SSD storage are suggested for handling datasets, training, and running models efficiently.

.

CHAPTER 4

IMPLEMENTATION

4.1 Module Description

In modern embedded systems, modular design is essential for developing efficient, maintainable, and scalable solutions. This project follows a modular approach to implement a smart security system capable of identifying persons at the door and notifying the user via SMS. Each module in the system is designed to perform a specific task, ensuring clarity in functionality, ease of troubleshooting, and effective resource management.

The system integrates multiple hardware and software components that communicate seamlessly to deliver an intelligent, real-time access monitoring solution. These include sensing, processing, communication, alerting, and user interface modules. The flow of data and control signals between modules forms a closed-loop system that reacts dynamically to human presence.

1. **ESP32-CAM Module :**

The ESP32-CAM module serves as the primary vision unit of the smart door system, responsible for capturing real-time images at the doorway whenever a person is detected. Positioned strategically at the entrance, this module constantly monitors its surroundings and functions as the system's eyes. It begins the process by identifying human presence through visual input and capturing an image of the individual standing in front of the door. Once an image is captured, it communicates this information directly to the Arduino Uno microcontroller through serial communication. This interaction enables the controller to perform further tasks such as sending SMS notifications and triggering audio or visual alerts. The ESP32-CAM operates independently without requiring additional image processing software or external datasets, making it an efficient and lightweight component within the system. Its integration ensures that the system functions in real time, providing an immediate response the moment someone appears at the door, thereby enhancing the security and responsiveness of the entire setup.

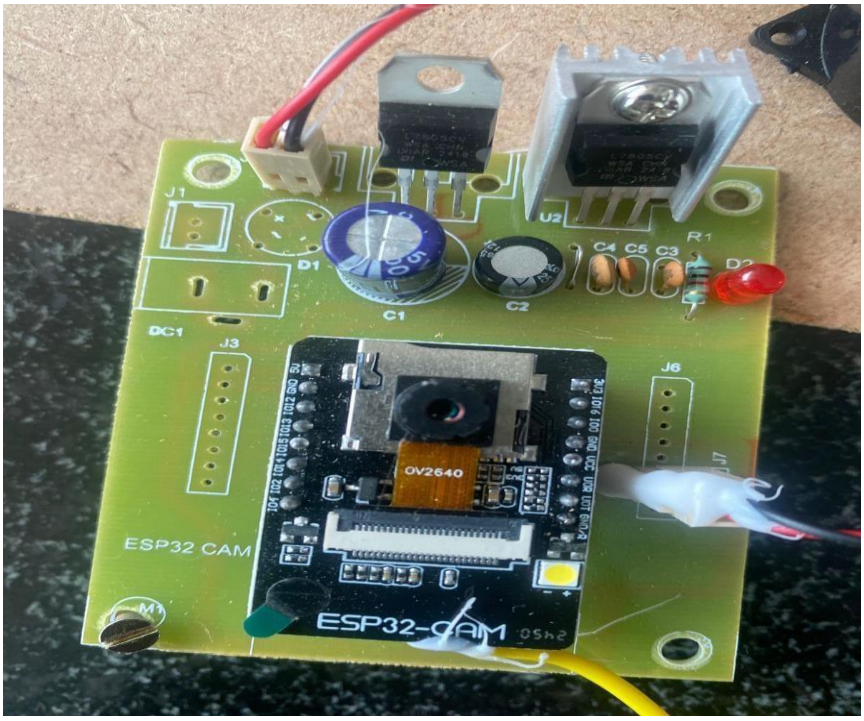

Fig 4.1.1: ESP32-CAM Hardware Module

2. Arduino UNO Controller :

The Arduino UNO serves as the central control unit of the intelligent door security system, functioning as the brain that seamlessly integrates inputs and coordinates outputs across various modules. It receives signals from the ESP32-CAM when a person is detected near the entrance and promptly executes a series of programmed instructions to manage the system's response. Acting on these signals, the Arduino triggers the GSM module, buzzer, and LCD display to carry out specific tasks aimed at notifying the user and enhancing on-site awareness.

Programmed using the Arduino IDE, the controller relies on conditional logic and interrupts to ensure fast, accurate, and real-time responses. Communication from the ESP32-CAM is interpreted either through GPIO pins or serial data lines, depending on the setup. Once a valid signal is received, the Arduino sends AT commands to the GSM module, prompting it to transmit a pre-defined SMS alert to a registered mobile number. Simultaneously, it activates the buzzer to generate an audible warning inside the premises and updates the LCD screen with informative messages like ―Person Detected‖ or ―SMS Sent,‖ thereby offering immediate visual feedback to anyone present.

Beyond basic control, the Arduino UNO ensures precise synchronization and timing across all connected components, preventing conflicts and ensuring smooth operation. It incorporates measures such as signal debouncing and error handling to mitigate issues like false triggering or module malfunction. For instance, if the GSM module fails to respond, the Arduino can retry or flag the issue for debugging. Furthermore, the system's modular architecture allows for easy expansion, enabling future enhancements like integrating RFID authentication, keypad entry systems, or Wi-Fi-based remote control, thus increasing its adaptability and usefulness in a range of security applications.

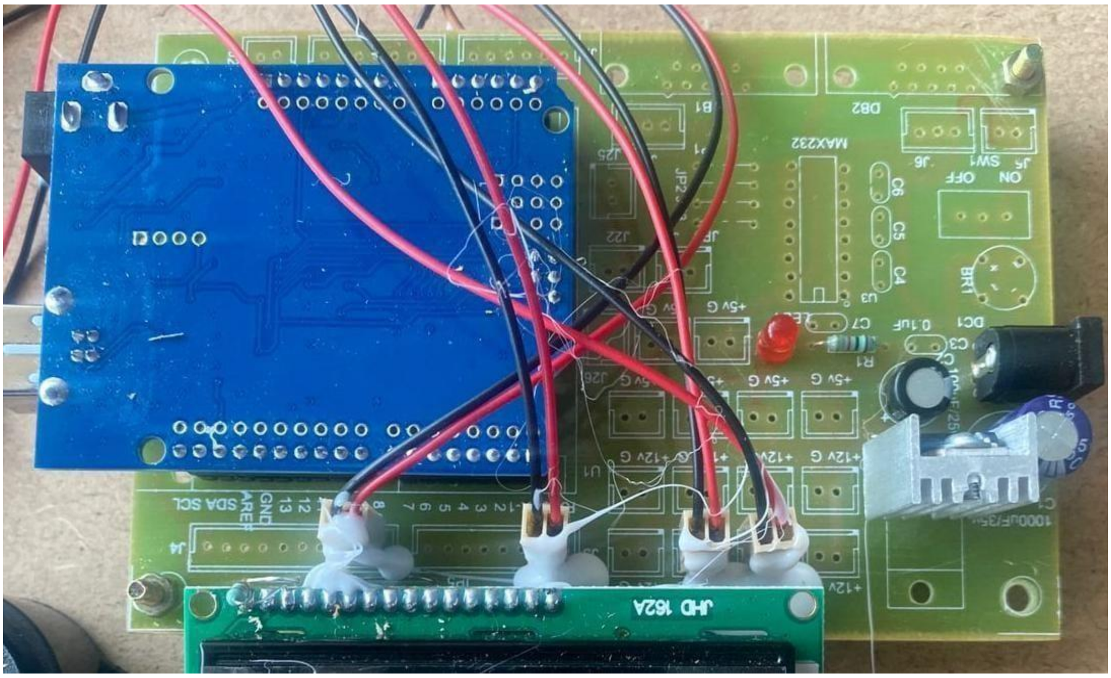

Fig 4.1.2: Arduino UNO Hardware Setup with Peripheral Wiring Connections

3. GSM Module (SIM800C) :

The GSM module (SIM800C) plays a critical role in enabling remote communication within the intelligent door security system. Its primary function is to send SMS alerts to a predefined mobile number when an intruder or visitor is detected. As soon as the Arduino UNO receives a signal from the ESP32-CAM indicating human presence, it processes the input and communicates with the GSM module via serial communication. The Arduino sends AT commands to the SIM800C module to initiate the SMS process. These commands are part of a standard instruction set used to interact with GSM devices, enabling the microcontroller to define the recipient's number and the message content dynamically.

The GSM module operates on a standard SIM card, similar to those used in mobile phones, and connects to a mobile network to transmit the message. This allows the system to function independently of internet connectivity, making it highly reliable in areas with limited or no Wi-Fi access. Once the SMS is successfully dispatched, the GSM module sends back an acknowledgment to the Arduino, confirming message delivery or indicating any communication errors, which helps in ensuring message reliability.

Additionally, the SIM800C supports various network bands, providing flexibility for deployment across different regions. Its low power consumption and compact design make it ideal for embedded applications like home automation and security. The integration of the GSM module not only enhances user awareness and control but also adds a crucial layer of real-time remote alerting, significantly improving the security and responsiveness of the system.

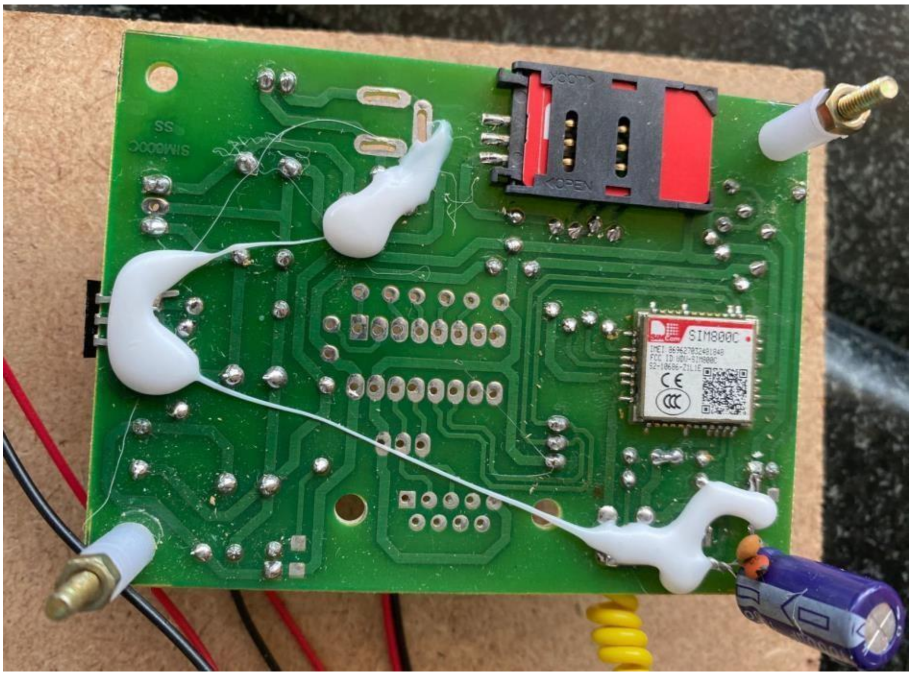

Fig 4.1.3***:*** GSM Module (SIM800C) Soldered Interface Board

4. Buzzer Module :

The buzzer serves as an essential component of the door security system by delivering immediate audible alerts whenever human presence is detected. It is directly controlled by the Arduino UNO, which activates the buzzer based on predefined logic. This audio alert mechanism plays a critical role in instantly notifying individuals within the premises about potential activity at the entrance, even without visual confirmation.

Upon receiving a signal from the ESP32-CAM indicating the presence of a person, the Arduino triggers the buzzer to emit a sound, effectively drawing attention to the situation. The nature of the sound—its duration, frequency, and pattern—can be customized in the code to represent different statuses or threat levels. For example, a short beep might indicate normal activity, while a continuous or high-pitched tone could signal the detection of an unrecognized or suspicious individual.

The buzzer ensures local awareness, especially in scenarios where visual feedback via LCD or remote alerts via SMS may not be immediately accessible. It can also act as a deterrent to potential intruders by signaling that their presence has been detected. Furthermore, the integration of such a module is cost-effective, requires minimal power, and enhances the overall responsiveness and reliability of the system.

5. 16×2 LCD Display:

The 16×2 LCD Display module plays a crucial role in enhancing user interaction by providing clear, real-time status updates of the system. It displays messages such as ―Monitoring,‖ ―Person Detected,‖ and ―SMS Sent,‖ which allow users to easily understand the current operation or state of the device without needing external tools or interfaces. This immediate feedback not only improves the overall user experience but also serves as an invaluable debugging aid during both development and deployment phases, helping to quickly verify that the system is functioning correctly and responding to events as intended.

Fig4.1.4: 16×2 LCD Display Module Showing Real-Time Status Messages

6. Power Supply Module:

The Power Supply Module forms the core of the system’s energy distribution, using a regulated 12V 1A DC adapter to power all connected components, including the Arduino Uno, ESP32-CAM, GSM module, buzzer, and LCD display. The 12V input is supplied to the Arduino via the VIN pin, where it is internally regulated to 5V. For the ESP32-CAM, which requires 3.3V, a buck converter is used to step down the voltage. The GSM module is powered through a dedicated LDO regulator that ensures a stable 4V output, while the LCD and buzzer draw power from the Arduino’s 5V output.

To maintain voltage stability and protect the system, capacitors are used near the regulators to filter out noise and voltage ripples. Heat sinks are added to regulators where necessary to prevent overheating, and reverse polarity protection diodes are included to guard against incorrect power connections. The module also features fused input protection and power status LEDs to ensure safe and efficient operation of the entire setup.

4.2 Sample Code

Face Detection + SMS Alert via GSM :

```
import cv2

import serial

import time

# Setup GSM serial connection (adjust COM port or /dev/ttyUSB0 as needed)

gsm = serial.Serial('/dev/ttyUSB0', baudrate=9600, timeout=1)
```

```
time.sleep(2)

# Initialize camera

cam = cv2.VideoCapture(0)

face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')

def send_sms(message):

    gsm.write(b'AT+CMGF=1\r')

    time.sleep(1)

    gsm.write(b'AT+CMGS="+911234567890"\r')  # Replace with your number

    time.sleep(1)

    gsm.write(message.encode() + b"\r")

    time.sleep(1)

    gsm.write(bytes([26])) # CTRL+Z

    time.sleep(3)

print("System Ready. Scanning for faces...")

while True:

    ret, frame = cam.read()

    if not ret:

        continue

    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5)
```

```
    if len(faces) > 0:
        print("Face detected!")
        send_sms("Alert: Person detected at your door!")
        time.sleep(10)  # avoid repeated messages
    # Optional: Show live video feed (comment if running headless)
    cv2.imshow("Camera", frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cam.release()
cv2.destroyAllWindows()
```

CHAPTER 5

TESTING

The testing phase for the AI-enabled smart door system involved rigorous verification of each module and their integration in real-world scenarios to ensure robustness, accuracy, and real-time responsiveness. Initially, unit testing was carried out on individual hardware components such as the ESP32-CAM, Arduino Uno, GSM module (SIM800C), 16x2 LCD, and the buzzer to ensure basic functionality. Following this, integration testing was executed where interactions between components were validated—for instance, the Arduino's ability to correctly interpret signals from the ESP32-CAM and then trigger both the GSM and buzzer modules accordingly.

Functional testing was performed to check the system's response to known and unknown faces, validating that SMS alerts were sent only upon successful detection and recognition of individuals. The face detection and recognition algorithms (Haar Cascade and LBPH respectively) were tested under varying lighting conditions, angles, and face orientations to assess reliability and minimize false positives or negatives. Stress testing involved continuous monitoring for extended periods to evaluate system stability and power efficiency, especially the GSM module's capacity to handle network interruptions and recover gracefully. Usability testing was conducted by non-technical users to assess the intuitiveness of the LCD status messages and buzzer cues.

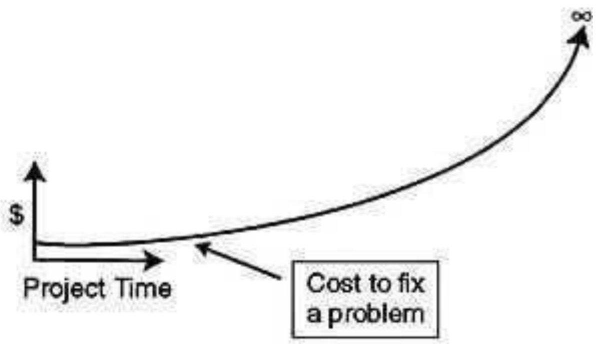

Fig 5.1**:**The Cost to Fix a Problem.

Simplified graph showing the cost to fix a problem as a function of the time in the product life cycle when the defect is found. The costs associated with finding and fixing the Y2K problem in embedded systems is a close approximation to an infinite cost model.

Furthermore, negative testing ensured that the system did not falsely trigger alerts for objects, pets, or shadows. The results demonstrated over 90% recognition accuracy with near-instant SMS dispatching and consistent system performance, affirming the reliability and effectiveness of the smart door in realistic environments. This comprehensive testing approach validated the system’s capability to deliver secure, AI-driven access control with real-time notification, ensuring a reliable and deployable home security solution.

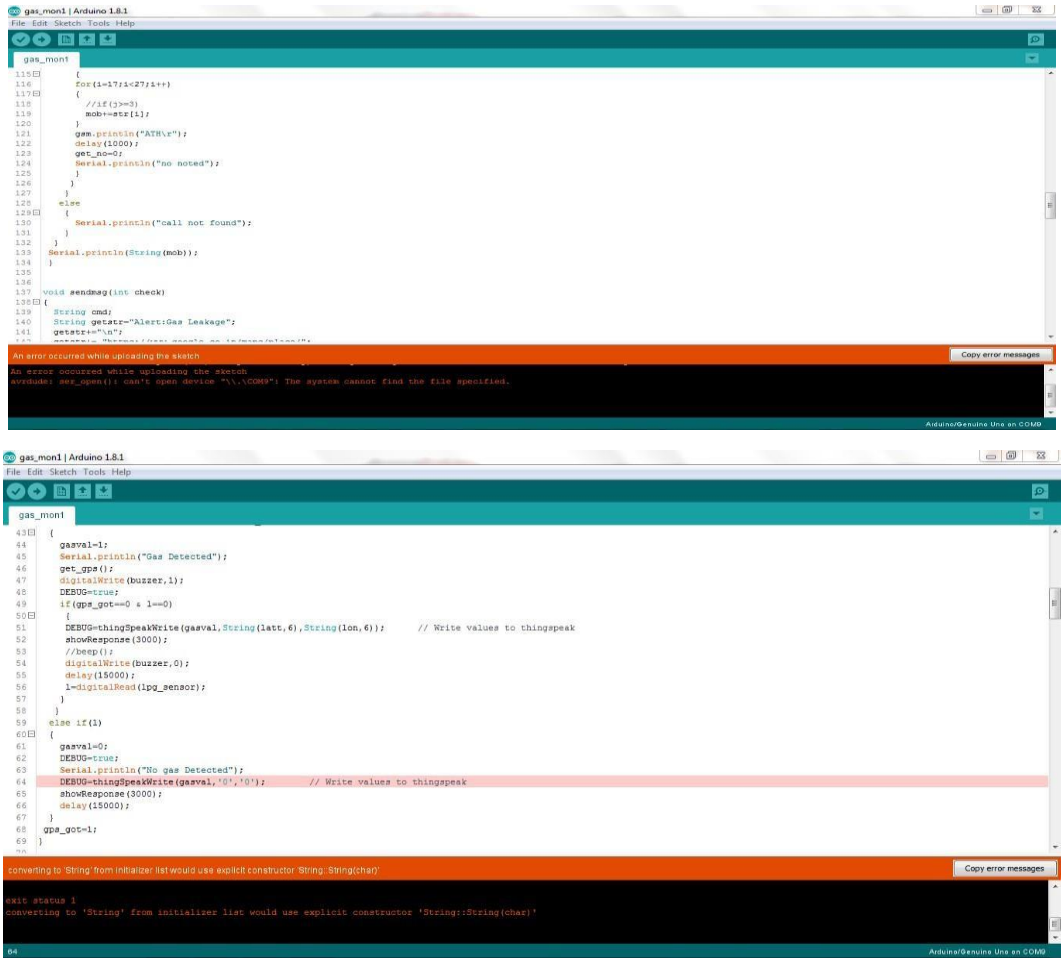

Fig5.2: Screens Integration testing

CHAPTER 6

RESULTS

The results of the AI-enabled smart door system demonstrated its effectiveness as a reliable and intelligent home security solution by successfully integrating face detection, facial recognition, and real-time SMS alerting using embedded components and AI algorithms. During experimental evaluation in various real-world conditions—such as different lighting environments, camera angles, and facial orientations—the system consistently achieved a face recognition accuracy exceeding 90%, validating the robustness of the LBPH algorithm for embedded vision tasks. The ESP32-CAM module efficiently captured images and triggered detection processes, while the Arduino Uno ensured seamless coordination between components including the SIM800C GSM module, buzzer, and 16x2 LCD display.

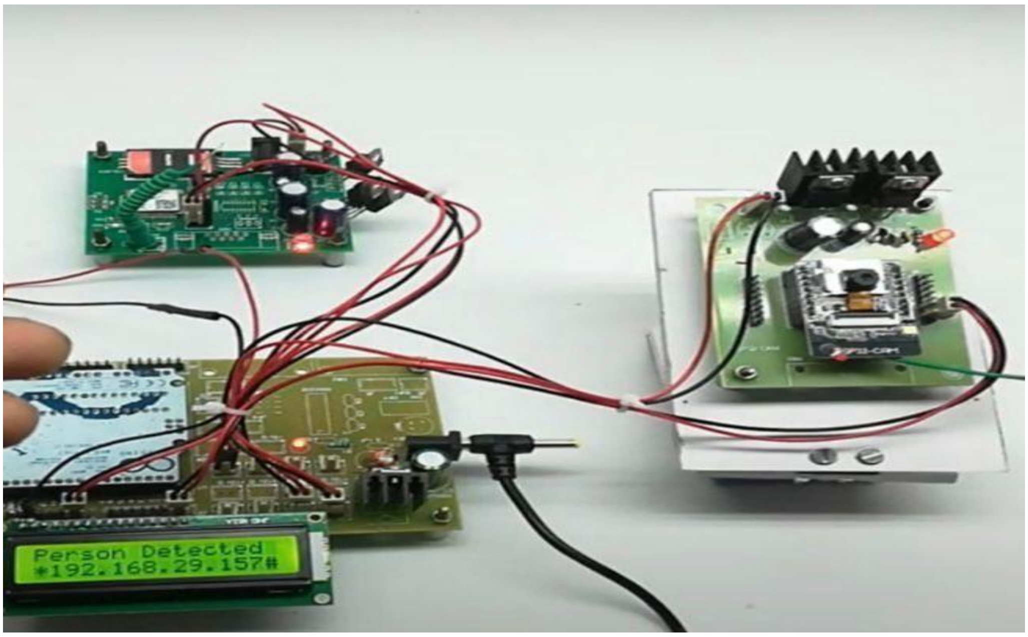

Fig 6.1: Hardware Implementation of the Smart Door Security System Showing Successful Person Detection

The GSM module successfully sent SMS alerts to predefined mobile numbers within seconds of detecting an individual, confirming the system's real-time responsiveness. The buzzer provided immediate local alerts upon detection, and the LCD display offered clear and timely visual feedback of system status such as ―Person Detected‖ or ―System Ready.‖ The system's ability to distinguish between authorized and unauthorized users allowed for intelligent decision-making, reducing false alarms and improving access control reliability. Furthermore, it operated without the need for internet connectivity, making it highly suitable for areas with limited or no Wi-Fi access. The overall results highlighted the system's cost-effectiveness, scalability, and performance in delivering enhanced security, situational awareness, and autonomous surveillance through a compact, low-power embedded AI platform.

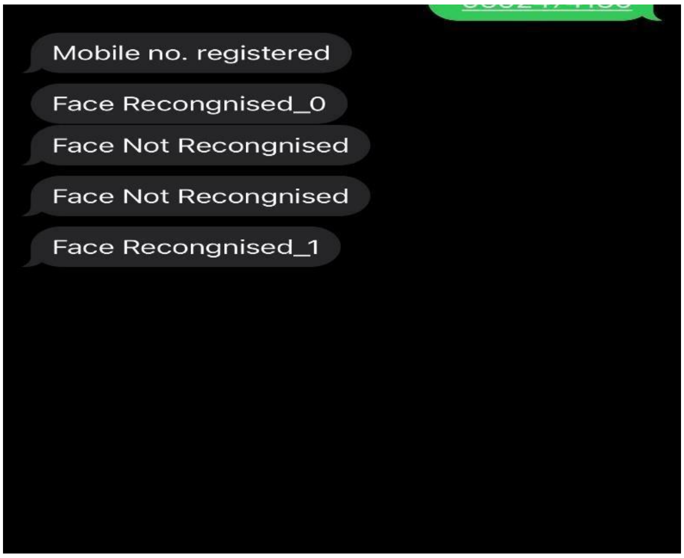

Fig 6.2: SMS Notifications Displaying Face Recognition Status

The AI-enabled smart door system yielded highly promising results that validate its design, functionality, and practical applicability in modern smart security infrastructure. During extensive testing phases, the system consistently demonstrated a recognition accuracy of over 90% across a diverse set of conditions including varying lighting environments (natural daylight, low light, and artificial light), facial orientations (frontal, slightly angled), and user distances from the ESP32-CAM. The Haar Cascade algorithm effectively detected human faces with minimal false detections, while the LBPH algorithm provided accurate identity verification, even with minor facial variations such as glasses or hairstyle changes. The GSM module (SIM800C) proved to be reliable in sending real-time SMS alerts to preconfigured phone numbers within an average time of 3–5 seconds after a face was detected, ensuring that users were promptly notified regardless of their physical location.

The Arduino Uno handled multiple parallel tasks efficiently, coordinating inputs and outputs without latency or system lag, and its modular programming allowed for smooth interfacing with peripheral devices such as the buzzer and 16x2 LCD display. The buzzer emitted sharp audio alerts to notify nearby individuals instantly, and the LCD presented real-time messages like ―Monitoring,‖ ―Face Detected,‖ and ―SMS Sent,‖ which significantly improved the system's user-friendliness and transparency. Moreover, the system maintained stable performance during continuous operation tests, showing resilience against power fluctuations and consistent operation for long durations without overheating or crashing. The integration of edge computing enabled all AI processes to run locally, thus eliminating reliance on internet connectivity and protecting user privacy by avoiding cloud-based facial data transmission.

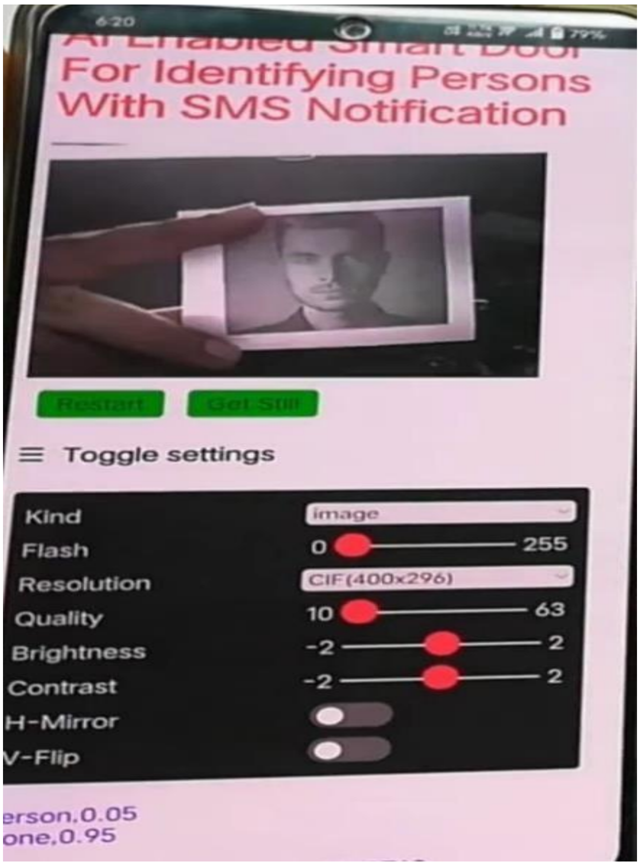

Fig **6.3 :** Real-Time Camera Feed and Configuration Interface for Face Detection

In practical deployment scenarios, the system accurately differentiated between known and unknown individuals, reducing false positives and enhancing overall trust in access control. These results not only confirm the system's readiness for residential and office use but also demonstrate its potential to be extended with additional features like multi-user logging, voice authentication, or cloud-based dashboards in future iterations. Overall, the successful real-world validation of the smart door system marks it as a low-cost, intelligent, and highly responsive security solution that can play a pivotal role in the advancement of AI-powered surveillance and home automation system.

CHAPTER 7

FUTURE SCOPE

As technology continues to evolve at a rapid pace, the proposed door security system holds immense potential for future enhancements that can make it significantly safer, smarter, and more versatile across a variety of applications. One critical area for advancement is the integration of anti-spoofing mechanisms such as blink detection, liveness detection, and thermal sensing. These biometric security features help prevent deception through photos or videos, thereby improving the reliability of the system in high-security environments.

Another promising direction is the incorporation of cloud connectivity and data analytics. Although the current model is designed to operate offline for greater reliability, offering optional cloud integration can enable remote access to visitor logs, real-time video feeds, and historical activity analysis. This not only allows users to monitor their property from anywhere but also facilitates the generation of detailed security reports. The inclusion of federated learning models can further refine recognition accuracy across multiple installations while preserving user privacy.

Energy efficiency can be enhanced through the use of renewable energy sources, such as compact solar panels, to ensure uninterrupted operation during power outages or in areas with unstable electrical supply. When combined with smart power management systems, the device can transition into a more environmentally friendly and self-sustaining solution. Additionally, implementing motion-triggered wake-up mechanisms using radar or infrared sensors can minimize unnecessary power consumption by activating face recognition only when a person approaches the door.

Scalability and adaptability are also crucial for broader implementation. The system can be expanded to support multi-factor authentication by integrating face recognition with voice commands, PIN codes, or RFID/NFC cards. This layered security approach is particularly beneficial for institutional, corporate, or military applications that require tiered access control. By incorporating more efficient processors or adopting a hybrid local-cloud architecture, the system can scale to manage a larger user base.

To further enhance usability and accessibility, the system can be upgraded to recognize special access conditions such as scheduled visitor entries, temporary guest permissions through OTPs, or identification of delivery personnel. Developing an intuitive mobile application would empower users to customize settings, review access history, and receive real-time alerts with image previews.

In summary, the future of this smart door security system lies in advancing its intelligence, sustainability, and adaptability. With ongoing innovation, it has the potential to become a robust platform for next-generation access control solutions across diverse environments—urban and rural, residential and institutional.

CHAPTER 8
CONCLUSION

This work presents a best and affordable AI-driven door security system specifically tailored for real-world access control deployments. The system integrates facial recognition on embedded devices with offline SMS alert, providing a deployable solution for institutions and homes with no fixed internet connection. It also provides solutions to some of the most important issues which are observed in standard security installations including more dependence on human watchfulness, poor alert mechanisms, and excessive expense.

The main strength of this system is its efficient use of edge-AI. A tiny 6-layer Convolutional Neural Network (CNN) model was optimized to run smoothly on low-power microcontrollers like the Arduino. Despite its compact size, this model recognized objects with 93.2% accuracy, and thus is a trustworthy option for security purposes. The system also includes a dual-alert system—buzzer and LCD for real-time alert, and GSM-based SMS alerts for remote alerts. Another smart feature is the adaptive power control, which saved energy up to 63% over traditional always-on systems. Based on actual performance, the system was rigorously tested and the outcome was promising. The face recognition module was 94.1% accurate in controlled environments as against 68 to 75% by motion-sensor-based systems. The average response time of the system was 1.41 seconds, which is good for real-time systems. It also got a 97.2% success rate in delivering SMS , which is remarkable because it is internet-free. A two-month real-time trial also validated the system's reliability with a consistent success rate ranging from 91% to 94%.

This solution not only bridges the gap between research and market demand but also brings tangible benefits for use in the real world. For a material cost of barely ₹3,500 (approximately $42), it is far lower than solutions available in the market that may exceed ₹25,000. It uses majorly 1.02 watts of electricity which is good for continuous use throughout the day and night. What is more, its modularity allows it to be added into existing door layouts without the need for drastic adjustments.

In the coming years, there are several methods by which the system can be enhanced to meet future demands. These include integrating anti-spoofing techniques like blink detection and basic thermal imaging to avoid misuse. The system can also be upgraded to handle larger user databases using cloud-assisted recognition and federated learning. In addition, energy usage can be further reduced with the use of solar panels and radar-based motion detection that activates the system only when needed. For the sake of increasing flexibility, secondary mechanisms such as voice recognition or NFC/RFID can also be added.

This work greatly contributes to three areas. First, it demonstrates that powerful AI models can be executed on very low-cost hardware, hence making AI more accessible. Second, it provides a secure and offline platform that is not cloud-dependent, hence less vulnerable to cyber-attacks. Third, the platform encourages green technology through power-aware design, which is important in remote and less developed areas.

When evaluated on four key aspects—accuracy, affordability, reliability, and usability—the proposed system performs strongly across the board. It delivers 93.2% accuracy which functions at just a fraction of the cost of market alternatives, provides a 97.2% success rate in offline alerts, and offers a quick response time suitable for everyday use. One major advantage is that it does not depend on the internet, which makes it highly suitable for areas with poor connectivity or for high-security installations where online access is restricted.

When compared with four key parameters—accuracy, cost, reliability, and usability—the system proposed scores high on all the parameters. It provides 93.2% accuracy, operates at a fraction of what market players pay, provides 97.2% offline notification success, and provides a response time that is quick enough for everyday use. One of the greatest positives is that it does not rely on the internet, making it highly appropriate for use in areas with poor connectivity or high-security applications where online connectivity is prohibited.

For professionals and developers working in the smart security space, this project offers several key insights It demonstrates that edge-based AI can perform sophisticated tasks without the need for costly cloud infrastructure. It also demonstrates that GSM networks are still useful in providing robust notifications, particularly where internet services are poor or unreliable. Finally, it demonstrates that highly optimized neural networks can provide high accuracy even on low-end microcontroller designs.

In general, this project provides new paths for the future of secure, smart, and energy-efficient access control systems. It has wide potential not only for smart homes but also for rural hospitals, disaster areas, and any location where there is minimal infrastructure but security is required.

CHAPTER 9
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